



Ramakrishna Mission Vivekananda Educational & Research Institute

Belur Math, Howrah, West Bengal
Department of Computer Science

MSc Big Data Analytics, II Semester, Final Examination

Course : **CS110: Data Structures and Algorithms**

Instructor : *Punyeshwarananda*

Student Id:

Student Signature:

Date: 17 May 2024

Time: 3 hrs

Max marks: 100

Answers should be clear, concise, and complete to earn the full allotted marks. All steps should be clearly stated while working out the problems.

First question is mandatory. Answer any 9 questions from the rest

1. (a) Give the prefix notation of this expression: $A - B/(C * D^E)$ (10)
(b) what is the worst case time complexity of merge sort?
(c) Give an example of a positive function $f(n)$ such that $f(n)$ is neither $O(n^3)$ nor $\Omega(n^3)$
(d) What are the minimum number of comparisons needed to find the smallest and largest integers from an array of 500 integers?
(e) What is the worst case time complexity of inserting n elements into an empty linked list, if the linked list needs to be maintained in sorted order?
(f) What is the data structure used to store nodes in breadth-first tree traversal?
(g) State whether the following statements are *True* or *False*
 - i. DFS is used to find the shortest distance in a graph G having unity weights
 - ii. The worst case running time of shell sort is sub-quadratic
 - iii. Hashing is a good choice for implementing sorted maps.
 - iv. The worst case running time of any comparison-based algorithm for sorting n elements $O(n \log n)$
2. Show that the bottom-up construction of a heap with n entries takes $O(n)$ time, assuming two keys can be compared in $O(1)$ time. (10)
3. Give the pseudocode for quicksort algorithm. What are its best and worst case performances and state when they occur. (10)
4. (a) The Master's theorem fails on this recurrence equation $T(n) = T(n/2) + n(2 - \cos n)$. Explain why? (5+5)
(b) A stack S contains N positive integers. Give a method to reverse the elements of the stack, you may only use any number of additional stacks. What is the time & space complexity of your method?
5. (a) Define heap data structure. Show that a heap T having n entries has height $h = \lfloor \log n \rfloor$ (6+4)
(b) Consider a max-heap, represented by the array 51, 30, 20, 10, 15, 16, 17, 8, 4. A new value 33 is inserted into the heap. What will be the order of elements in the array after insertion?
6. (a) What are the key features of a Hash function? (3+7)
(b) The following keys 45, 13, 34, 67, 23, 74 are hashed into a hash table of size 10 using $h(k) = k \% 10$. Give the sequence of the elements stored in the hash table where collisions are resolved using (a) linear probing and (b) quadratic probing (ie. recursively check for locations with indices $(h(k) + i^2) \% 10$, for $i = 0, 1, 2, \dots$)

7. (a) What are the best case and worst-case times for putting n entries in an initially empty hash table, with collisions resolved by chaining? Describe the circumstances under which they occur. (4+6)
- (b) What does the Python code shown in Listing (a) do? What is the time complexity of this code? Can it be improved, if so, how?
8. Let G be an undirected graph whose vertices are the integers 1 through 8, and its adjacency list is given in Table (b). (10)
- i). Draw G . ii). Starting from vertex 1, give the sequence of vertices of G visited using DFS and BFS traversals
9. (a) What are the distinct features of Bellman-Ford Algorithm? Using this algorithm find the shortest paths from node 3 to other nodes in Fig. (c) given below. (6+4)
- (b) The number of operations executed by algorithms P and Q are $8n \log n$ and $2n^2$, respectively. Determine n_0 such that P is better than Q for $n \geq n_0$
10. (a) You are asked to sort the pixels of a 8-bit grayscale image of size 100×100 . Describe a method to sort them optimally. Analyse its time-complexity. (5+5)
- (b) Let S be a sequence of n elements, each equal to 0 or 1. How long will it take to sort S by: mergesort, quicksort, selection sort and insertion sort?
11. (a) Describe how to implement a stack ADT given a priority queue ADT to you. (4+6)
- (b) What is the time-complexity of the following operations:
- i). Towers of Hanoi with n disks ii). Heap sort on n items iii). Matrix addition ($n \times n$ matrices) iv). DFS on $G(V, E)$ v). Kruskal's MST algorithm on $G(V, E)$ vi). Binary search on n sorted items

Listing (a)

```
def func1(arr):
    for j in range(len(arr)):
        for k in range(j+1, len(arr)):
            if arr[j] == arr[k]:
                return False
    return True
```

Table (b)

Vertex	Adjacent Vertices
1	(2, 3, 4)
2	(1, 3, 4)
3	(1, 2, 4)
4	(1, 2, 3, 6)
5	(6, 7, 8)
6	(4, 5, 7)
7	(5, 6, 8)
8	(5, 7)

Fig. (c)

